

Daniel R. Russell

Founder-operator and applied ML/DS engineer building **SNPTX**, a governed execution layer for multi-modal biomedical ML spanning eight modality families across 46 datasets (ChEMBL, Hetionet, Open Targets, TCGA, GTEx, MIMIC-IV, AlphaFold, Visium, DepMap, BC5CDR, eICU). Four operational surfaces: an autonomous experimentation engine (Gaussian-process surrogates with Expected-Improvement + Value-of-Information acquisition and SPRT stopping, validated at 1,037 experiments per hour), a contract-validated extension runner, a cross-lab benchmark surface with calibration-corrected leaderboards, and a deployment pathway with FastAPI, RBAC, and hash-chained audit. PyTorch model layer covers graph neural networks, diffusion, and transformer-based NLP; LangChain LLM-agent tooling on top. Supported by the **Harvard Innovation Labs** while completing the Harvard ALM in Data Science & Artificial Intelligence (4.00 GPA). Previously founded **FIVSS**, acquired by **Booking Holdings**.

Recent Role • Founder & Systems Engineer, SNPTX • ML/DL framework for biomedical discovery (GNNs, DL, LLM/NLP, Autonomous Learning System) • **Harvard Innovation Labs**

Technical Focus • PyTorch • GNNs • Deep Learning • NLP/LLM workflows • Snakemake/DVC/MLflow • AWS/GCP Parallel GPUs and GPU clusters

Postgraduate • Harvard ALM in Data Science & Artificial Intelligence (HES '27) • MITx MicroMasters, Statistics & Data Science ('24), MIT Institute for Data, Systems and Society (IDSS)

Prior founder exit • FIVSS → Booking Holdings

Email • dan@snptx.ai • dr508@g.harvard.edu

Web • [snptx-academic](https://snptx-academic.github.io) • [github @snptx1](https://github.com/snptx1)



INSTITUTIONS HARVARD UNIVERSITY • HARVARD INNOVATION LABS • MASSACHUSETTS INSTITUTE OF TECHNOLOGY (IDSS) • SNPTX

At a glance

A quick read on the work: multi-modal biomedical ML across 8 modality families / 46 datasets, autonomous GP-guided experimentation (1,037 exp/hr, 104 discoveries/campaign, surrogate ECE = 0.055), calibration-corrected cross-lab benchmarks, LLM-agent tooling, and a FastAPI + RBAC deployment surface - all under a governed runtime.

ACADEMIC SIGNAL

Harvard + MITx

Harvard ALM Data Science & AI, **4.00 GPA**; MITx MicroMasters, Statistics & Data Science.

COMPANY BUILDING

SNPTX + FIVSS

Active biomedical ML systems company; prior founder exit to Booking Holdings.

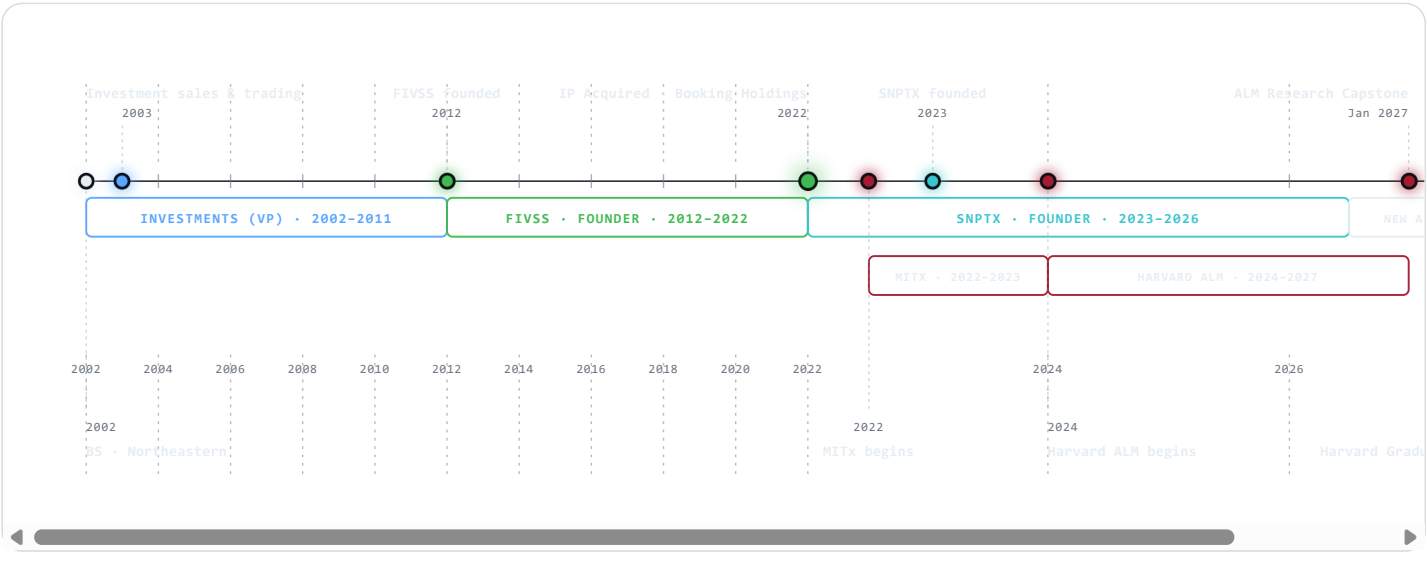
ML/DL SYSTEMS

GNNs + DAGs

GNNs, CNNs, VAEs, GANs, diffusion models, graph ML, Agentic RL, Bayesian inference and optimization, Snakemake/DVC/MLflow pipelines.

Career arc

A compressed early-career context followed by a deliberately recent-weighted founder, postgraduate, and applied-ML systems arc. The axis is intentionally non-linear: 2002–2022 occupies the left half, while the recent founder, postgraduate, and applied-ML period receives the full right half.



Professional experience

Reverse chronological. Founder-operator roles with full ownership of product, technology, and commercial workstreams.

SNPTX · Founder & Systems Engineer

HARVARD INNOVATION LAB

2022 – PRESENT

SNPTX · deterministic ML orchestration for biomedical discovery · supported by the [Harvard Innovation Labs](#) · Cambridge, MA

Founded and single-handedly built SNPTX, a governed execution layer for biomedical ML spanning eight modality families across 46 datasets. Four operational surfaces (autonomous experimentation via `ExperimentEngine` with GP surrogates, EI + VoI acquisition, SPRT stopping, and DuckDB catalog; a contract-validated extension runner; a deployment surface with FastAPI inference, RBAC, HMAC-signed tokens, hash-chained audit at 21 CFR Part 11 posture; and a cross-lab benchmark layer with calibration-corrected leaderboards) sit on top of a Snakemake / DVC / MLflow spine. Reported numbers are bounded to declared configurations, datasets, and seeds; nothing on the public site is a clinical-performance claim.

- Shipped a five-minute end-to-end reproduction repository ([snptx-repro-chembl](#)) of a ChEMBL bioactivity graph convolutional network baseline — deterministic at seed=42, `accuracy = 0.9851` on CPU, 98.0% reproducibility under fixed seeds. Companion reproduction ([snptx-repro-deepdds](#)) of the DeepDDS drug-synergy graph-attention model runs under the same deterministic harness — the trust artifact for external technical review of the scientific-AI pipeline.
- Built the `ExperimentEngine` autonomous-experimentation surface: Gaussian-process surrogates with Expected-Improvement + Value-of-Information acquisition, sequential-probability-ratio-test stopping ($\alpha = \beta = 0.05$), and a DuckDB experiment catalog. Validated at 1,037 experiments per hour and 104 discoveries per 1,100-experiment campaign (surrogate ECE = 0.055) — a working answer to the JD's design-of-experiments, Bayesian, calibration, and surrogate-model requirements.
- Integrated eight biomedical modality families across 46 datasets (ChEMBL, Hetionet, Open Targets, TCGA / TCGA-PanCan, GTEx, MIMIC-IV, AlphaFold, DepMap, Visium, BC5CDR, PubMed-class corpora) under a shared evaluation contract; benchmark surface reports 99.3% imaging, 92.8% omics, 98.5% drug-GCN, 97.5% ChEMBL antibacterial 5-fold CV, and 93.0% Visium spatial-transcriptomics fusion on declared configurations. Knowledge-graph work grounded in Hetionet + Open Targets with NetworkX / Neo4j analytics; biomedical NER on BC5CDR via BiLSTM-CRF and BERT-family transformers.
- Architected the full stack solo: Snakemake DAGs, DVC content-hashed data, MLflow experiment tracking, PyTorch model layer (graph neural networks, diffusion models, transformer-based NLP), LangChain LLM-agent tooling, FastAPI inference with RBAC + hash-chained audit (21 CFR Part 11 posture), containerized runtime, and CI-enforced reproducibility gates (1,357 tests, 0 Ruff warnings, 0 Pyright warnings).
- Concurrently completing the Harvard ALM in Data Science & Artificial Intelligence (4.00 GPA), routing graduate final projects into SNPTX as bounded reproducibility artifacts — DeepDDS graph-attention drug-synergy reproduction (CSCI E-104, Djordjevic), eICU ICU-mortality forecasting under VAR / dynamic-factor / state-space models (CSCI E-116), ChEMBL32 graph analytics with NetworkX and Neo4j (CSCI E-108, Elston — assigned capstone advisor), BC5CDR biomedical NER (CSCI E-89b, Kurochkin). Prior MITx MicroMasters in Statistics & Data Science (Rigollet, Tsitsiklis, Uhler, Barzilay, Jaakkola).

Snakemake

DVC

MLflow

PyTorch

Graph Neural Nets

Diffusion Models

LangChain / LLM Agents

Python

Docker

CI / Reproducibility Gates

Content-Hashed Data

EVIDENCE

[SNPTX-ACADEMIC](#)

[REPRO-CHEMBL](#)

[REPRO-DEEPDDS](#)

[GITHUB](#)

FIVSS · Founder

IP ACQUIRED · BOOKING HOLDINGS

2012 – 2022 · ACQUIRED

FIVSS · predictive-pricing platform for five-star hotel-suite inventory · [acquired by Booking Holdings](#) (NASDAQ: BKNG; parent of Booking.com, Priceline, Kayak, OpenTable)

Founded and led FIVSS through a ten-year build, from an original insight — that the secondary market for luxury suite inventory was systematically mispriced — to a proprietary ML pricing engine, a live transactional marketplace, and ultimately a strategic acquisition by [Booking Holdings](#). Operated as sole founder for the majority of the company's lifespan, owning every function from model design to legal-entity formation to buyer negotiation.

- ▶ **Product thesis:** Identified persistent inefficiencies in five-star suite pricing across metropolitan and resort markets. Built proprietary demand-forecasting and price-discovery models that flagged underpriced inventory in near-real-time for secondary-market resale.
- ▶ **Data & ML:** Designed and ran ingestion pipelines against rate-source APIs and structured scrapers; normalized and warehoused millions of price observations in SQL. Trained gradient-boosted and time-series demand models that generated per-suite fair-value estimates and rank-ordered opportunities by expected margin.
- ▶ **Platform:** Built the full transactional web application — inventory dashboard, buyer-facing UI, payments integration, and back-office reconciliation — as sole engineer for the founding stretch.
- ▶ **Commercial operations:** Formed the operating entity, negotiated supplier and partner agreements, ran the day-to-day P&L, and grew the business from bootstrapped prototype to a cash-flow-generating operator with recurring inventory relationships.
- ▶ **Exit:** Led the buyer-side workstream through diligence and close, culminating in the 2022 acquisition by Booking Holdings.

Predictive Pricing

Demand Forecasting

Gradient Boosting

Time-Series Models

SQL Warehousing

Python / ETL

Web Platform

Payments

Marketplace Ops

M&A Negotiation

Investment Sales & Trading · Vice President

PRE-2012

VP-level sales-and-trading roles in banking (Morgan Stanley, Merrill Lynch, etc.) including client coverage and market-facing execution in a large-account, high-throughput trading environment — the quantitative, commercial, and pressure-context substrate underneath the subsequent founder chapters.

SNPTX in brief

Public inspection surface for the current build. Every cell links to an artifact designed to be reviewed end-to-end without follow-up.

Deterministic harness

Seed pinned at [seed=42](#) across Python, NumPy, PyTorch, and CUDA. Environment locked via [pyproject.toml](#) plus lock file.

[METHODOLOGY](#) · [REPRODUCIBILITY](#) →

Five-minute reproduction

Carved-out repository reproduces the ChEMBL bioactivity GCN baseline end to end on commodity CPU in roughly five minutes, with stated headline metric.

[SNPTX-REPRO-CHEMBL](#) →

Inspection orientation

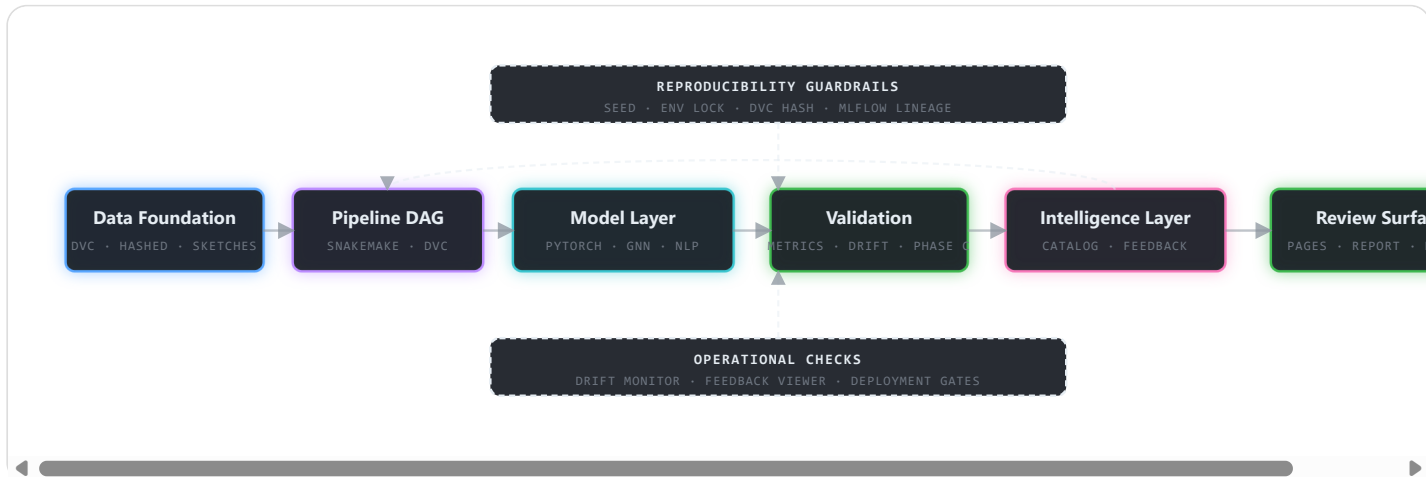
Short orientation page intended for senior technical readers, with three reading paths and an inspection-invariants panel.

[SNPTX-ACADEMIC](#) →

Methodology surface

Architecture, methodology, and references pages organize the design, the reproducibility controls, and the literature the work draws on.

[ARCHITECTURE](#) →



Selected technical work

Concrete, inspectable artifacts drawn from the SNPTX build and the concurrent Harvard graduate program. Each is designed to be reviewed end-to-end.

Five-minute reproduction repo

End-to-end reproduction of a ChEMBL bioactivity GCN baseline from a clean clone on commodity CPU. Stated headline metric, pinned environment, deterministic seed.

[reproduces in ~5 min · CPU-only](#)

[SNPTX-REPRO-CHEMBL](#) →

DeepDDS drug-synergy reproduction

Graph-attention drug-synergy model reproduced under the SNPTX deterministic harness. Full provenance and environment lock; served as the CSCI E-104 final project.

[A / 1.0000 raw score](#)

[SNPTX-REPRO-DEEPDDS](#) →

ICU mortality forecasting (eICU)

Temporal early-window ICU mortality forecasting on the eICU Collaborative Research Database, packaged as a self-contained reproducible notebook with dual real-data and synthetic-fallback execution paths.

[A · CSCI E-116 final](#)

[NOTEBOOK](#) →

ChEMBL32 bioactivity mining

Graph-analytic exploration of medicinal-chemistry data across ChEMBL32; scaled

similarity, clustering, and recommender workflows applied to real bioactivity records.

[A · CSCI E-108 final](#)

[NOTEBOOK →](#)

Education

Full academic detail — instructor-level coursework matrix, per-course raw scores, and named faculty references — available in the [detailed academic transcript](#).

ALM, Data Science & Artificial Intelligence

Harvard University (HES '27)

GPA 4.00 / 4.00 across all completed coursework. Capstone: deterministic reproducibility and provenance harness for multi-modal biomedical ML, advised by Stephen Elston, PhD.

MicroMasters, Statistics & Data Science

MIT Institute for Data, Systems & Society (IDSS) · MITx · 2022–2024

Full program awarded 2024. Rigorous probability, mathematical statistics, machine learning, and applied data-analysis sequence including the proctored capstone exam.

BS, Business Administration · Finance & International Business

Northeastern University

Undergraduate foundation in business administration, finance, and quantitative business analysis that complements the later founder/operator chapters.

GRADUATE COURSEWORK ([HARVARD ALM](#) · [MITX MICROMASTERS](#))

Advanced Deep Learning

Dynamic Modeling & Forecasting

Data Mining, Discovery & Exploration

Natural Language Processing

Epidemiology & Biostatistics

Foundations of Data Science & Engineering

Robotics, Autonomous Vehicles & AI

Machine Learning (MITx 6.86x)

Data Analysis: Modeling & Computation (6.419x)

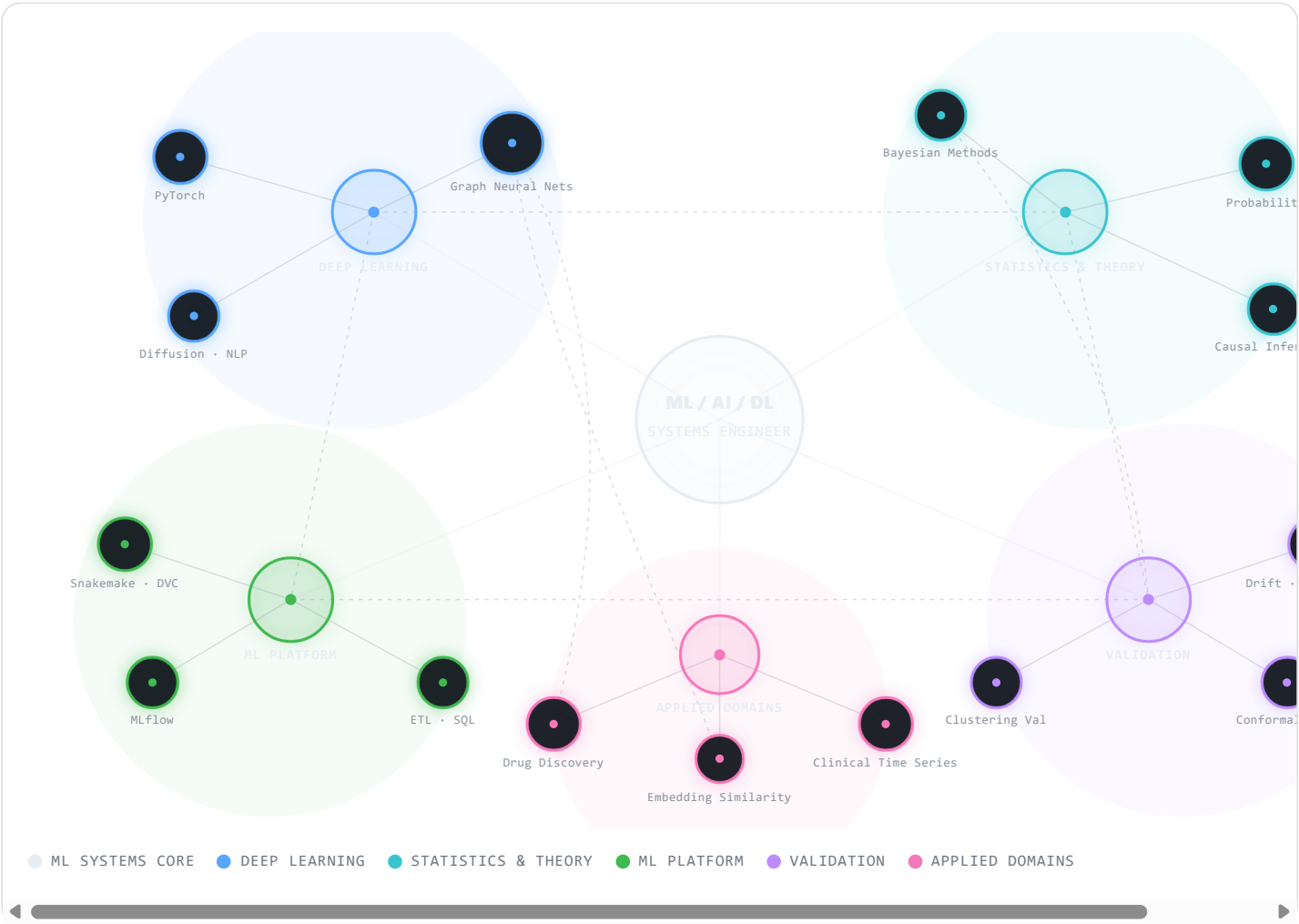
Probability (6.431x)

Fundamentals of Statistics (18.6501x)

MicroMasters Capstone Exam

Applied Capabilities

Hands-on capabilities organized by dimension — deep learning, statistics and theory, ML platform, validation, and applied domains. This is the working stack: what gets built, shipped, and validated on live engagements. The complementary *Foundational Theory* section below shows the concept toolkit that grounds this work.



COMMERCIAL & OPERATING

- Founder / Sole Operator
- M&A · Buyer-Side Diligence
- Entity Formation
- P&L Ownership
- Technical Writing & Reviewer Communication

LANGUAGES & TOOLCHAIN

- Python
- SQL (MySQL / Postgres)
- R
- Docker
- AWS / GCP
- Neo4j · NetworkX
- Git / GitHub

Foundational Theory

A map of the machine learning theory and deep learning architectures brought to every engagement — anchored in probability and mathematical statistics, extended through classical ML, deep learning, sequence and language models, reinforcement learning, and applied AI.

DEEP LEARNING & ARCHITECTURES

MLPs & backpropagation

CNNs & computer vision

Autoencoders & VAEs

GANs

Diffusion models

Graph neural networks

MATHEMATICS FOR ML & AI

Probability theory

Mathematical statistics

Bayesian inference

Hypothesis testing & LRT

Multiple testing & FDR

Causal & epi inference

GLMs & regression

MACHINE LEARNING & OPTIMIZATION

SGD / Adam / RMSProp

Bayesian optimization

SVM & kernel methods

Tree ensembles & boosting

Clustering & GMM-EM

PCA / t-SNE / UMAP

Regularization theory

REINFORCEMENT LEARNING

MDPs & Markov property

Bellman & value iteration

Q-learning & optimal policy

Policy gradient & actor-critic

Model-free vs model-based

RL for robotics & planning

Agentic RL & tool use

SEQUENCE, LANGUAGE & NLP

RNN / LSTM / BiLSTM-CRF

Attention & transformers

Word2Vec / GloVe embeddings

Tokenization (BPE / WP)

LDA / NMF / topic models

LLM agents & RAG

GRAPHS & RETRIEVAL SYSTEMS

Graph centrality (PageRank)

Community detection

Similarity search & LSH

Streaming & sketching

Recommender systems

Vector DBs & embeddings

TIME SERIES & DYNAMICS

ARMA / ARIMA / VAR

Cointegration & VECM

GARCH & volatility

State-space & Kalman

Dynamic factor models

Gaussian processes

APPLIED & SCIENTIFIC AI

Calibration & ECE

Conformal prediction

Drift & covariate shift

Robotics & SLAM

Physics-informed NNs

ML for drug discovery

References

Academic references drawn from graduate courses in which the final project fed directly into SNPTX. Each reference's scope is bounded to that course and its artifact.

Zoran B. Djordjević, PhD

REFERENCE

Harvard University · CSCI E-104 Advanced Deep Learning · Spring 2026 · A / 1.0000

Advanced deep-learning execution, graph-attention drug-synergy reproduction, and extension of course-level work into a deterministic SNPTX benchmark surface. Contact: zdjordj@fas.harvard.edu.

Stephen Elston, PhD

REFERENCE · CAPSTONE ADVISOR

Harvard University · CSCI E-108 Data Mining · Fall 2025 · A · assigned ALM capstone advisor

ChEMBL32 bioactivity mining and graph analytics, continuation into SNPTX seed datasets, and capstone supervision context. Contact: stephen_elston@g.harvard.edu.

Dmitry V. Kurochkin, PhD

REFERENCE

Harvard University · CSCI E-89b NLP · Fall 2025 · A

Biomedical NLP execution and hybrid statistical / neural NER over the BC5CDR corpus — the upstream text-processing components that now feed SNPTX. Contact: dkurochkin@fas.harvard.edu.